A. Correlation Magnitude vs Accumulation Length

This figure studies how coherent correlation accumulation evolves under different Doppler frequencies.

The received sequence is modeled using a deterministic Doppler-induced phase rotation:

$$r[n] = s[n]e^{\{j\ \omega n\}}$$

where:

- (s[n]) is the known reference sequence,
- (\omega) represents the Doppler-induced phase rotation rate.

The accumulated correlation magnitude is computed as:

C(L) = | \sum_{n=0}^{L-1} r[n] s^*[n] |

and normalized with respect to its maximum value for visualization consistency across Doppler conditions.

The figure illustrates how correlation evolves as the accumulation length (L) increases.

At low Doppler values:

- phase rotation progresses slowly,
- correlation terms remain coherent over longer intervals,
- and accumulation continues to grow constructively.

As Doppler increases:

- phase rotation becomes more rapid,
- coherent addition weakens earlier,
- and the accumulation curve saturates more quickly.

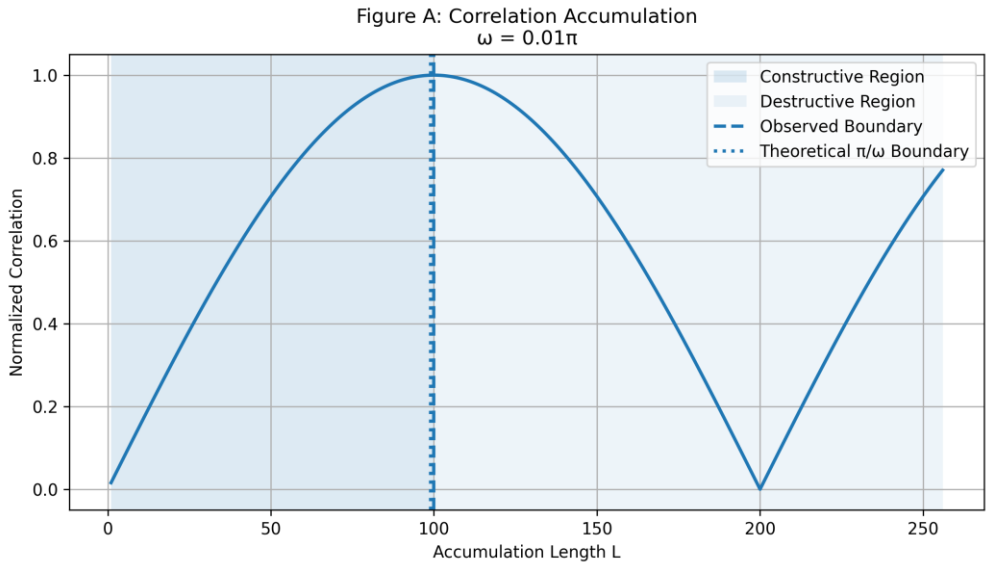
This behavior reveals the existence of an effective coherence-support region beyond which additional accumulation provides diminishing benefit.

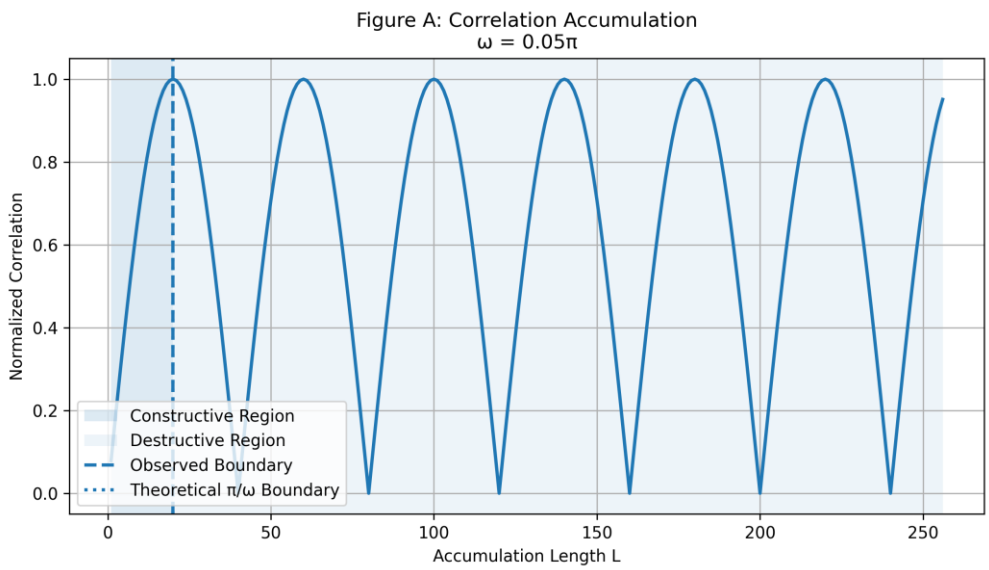
The shaded constructive region represents the interval where coherent accumulation remains beneficial, while the destructive region corresponds to accumulation lengths where phase misalignment increasingly reduces incremental contribution.

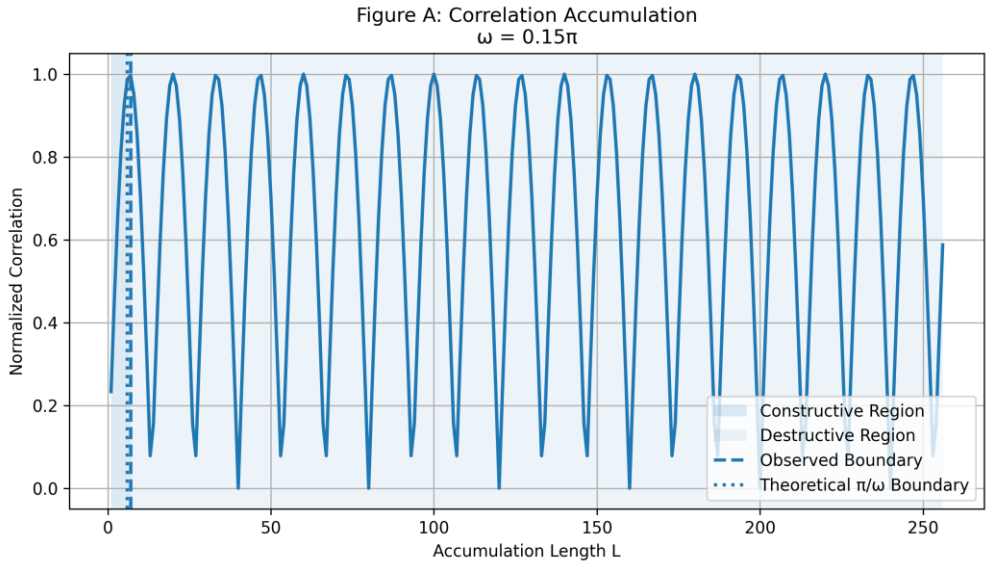
An important observation from the figure is that the effective coherence width decreases as Doppler increases, suggesting that optimal accumulation length should adapt dynamically to channel time variation.

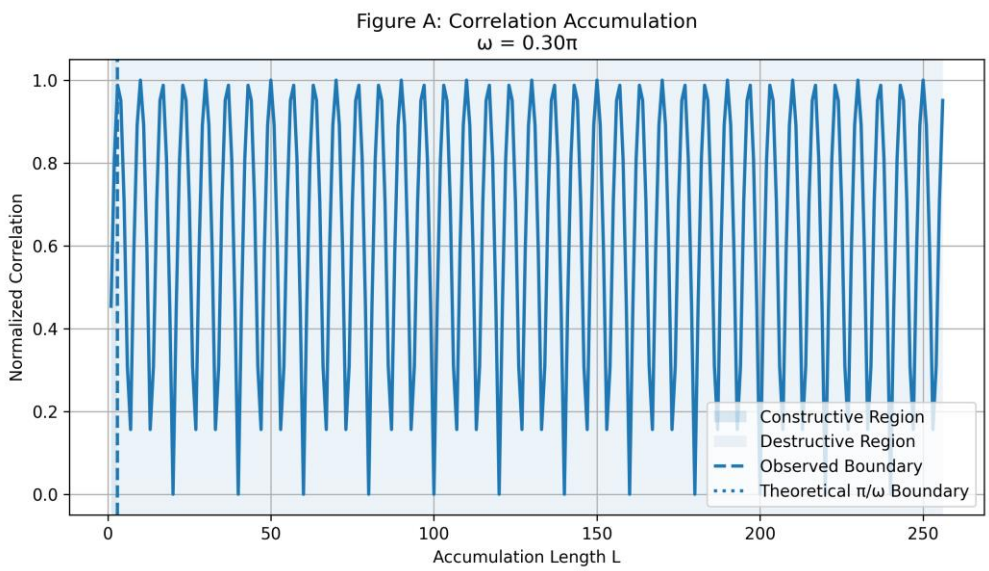
This provides intuition for:

- Doppler-aware coherent integration,
- adaptive receiver accumulation control,
- synchronization window selection,
- and intelligent PHY-layer adaptation strategies.









B. Incremental Gain vs Accumulation Length

While correlation magnitude provides a global view of coherent accumulation behavior, it does not directly reveal how much each additional accumulated sample contributes locally to the overall correlation strength.

To better understand this behavior, the incremental gain is computed as the change in normalized correlation magnitude between consecutive accumulation lengths.

Let the normalized accumulated correlation magnitude be represented as:

```
C_norm(L) = C(L) / max(C(L))
```

The incremental gain is then defined as:

```
g(L) = C_norm(L) - C_norm(L-1)
```

This quantity measures the additional contribution obtained by extending the accumulation window from (L-1) to (L).

The resulting figure provides a local view of coherent accumulation dynamics under different Doppler conditions.

At low Doppler:

- incremental gain remains positive over a longer interval,
- indicating that newly accumulated samples continue contributing coherently.

As Doppler increases:

- the incremental gain decays more rapidly,
- approaches zero earlier,
- and may eventually become negative due to phase misalignment.

The transition point where incremental gain approaches zero represents an approximate coherence-support boundary. Beyond this point, additional accumulation provides diminishing benefit and may become increasingly destructive.

This interpretation enables separation of the accumulation process into:

- constructive accumulation regions,
- and destructive accumulation regions.

Unlike global correlation magnitude alone, incremental gain directly exposes the local coherence evolution of the received signal and provides a physically meaningful criterion for adaptive accumulation control.

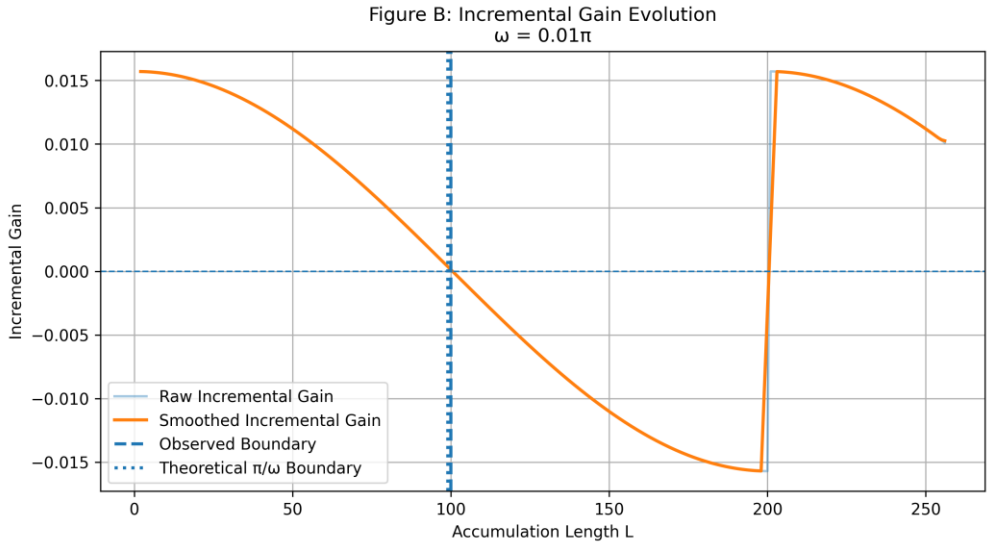
The figure also illustrates that:

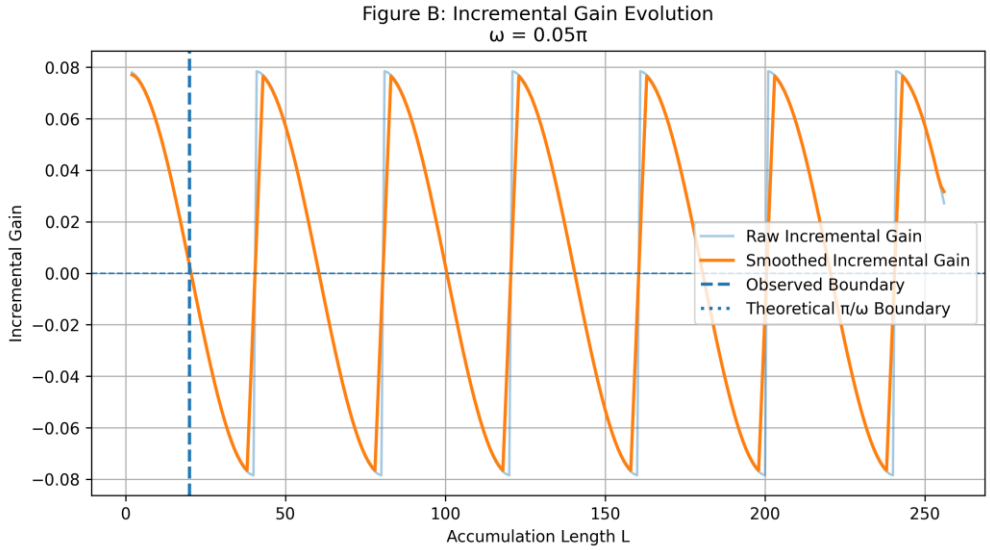
- higher Doppler reduces useful accumulation support,
- coherent integration efficiency decreases with increasing phase rotation rate,
- and accumulation length should ideally adapt to channel time variation.

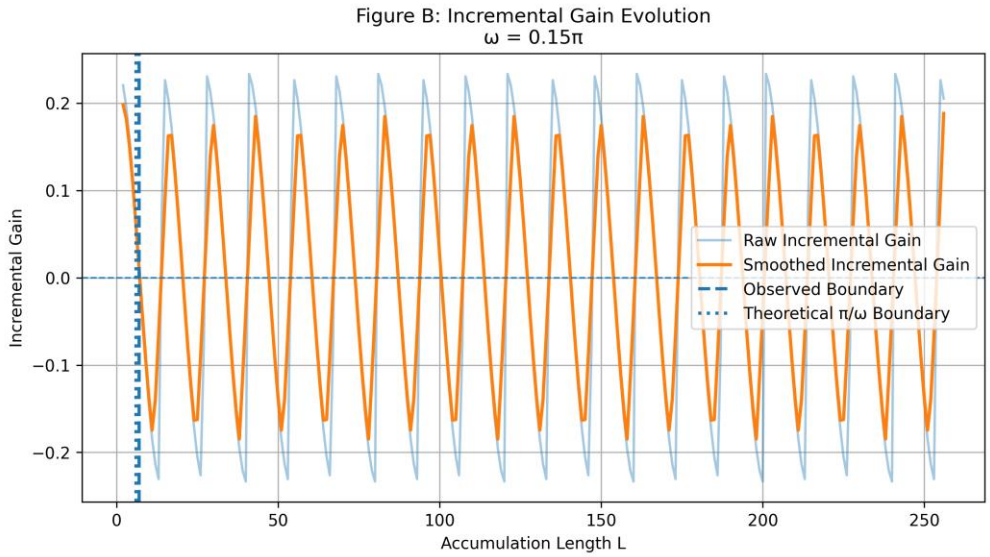
This type of analysis can be useful for:

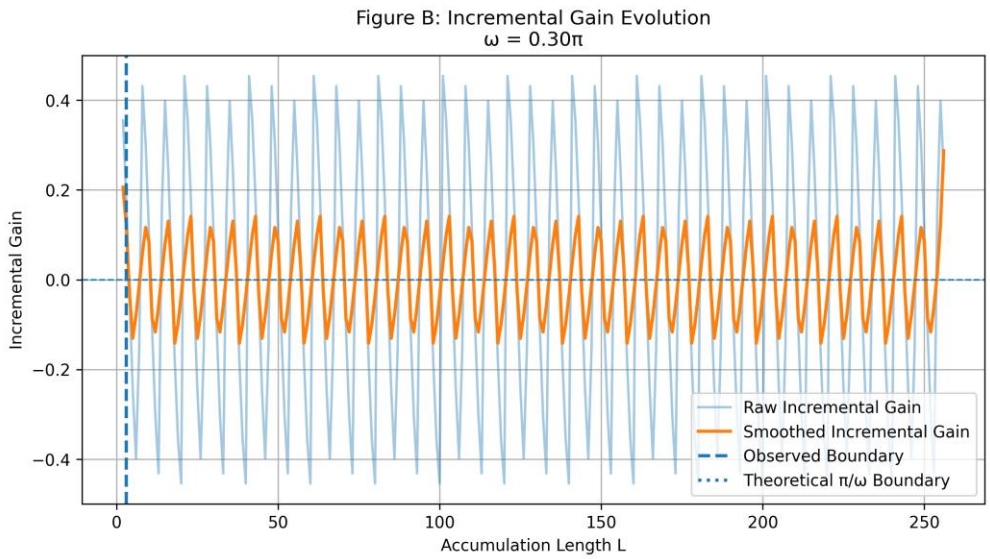
- adaptive coherent integration,
- synchronization/search-window optimization,
- pilot accumulation control,

- Doppler-aware receiver adaptation,
- and intelligent PHY-layer decision frameworks.









C. Coherence Width vs Doppler

The previous figures showed that coherent accumulation becomes increasingly limited as Doppler-induced phase rotation grows. To quantify this behavior, an effective coherence-support width is estimated for each Doppler condition.

The coherence width represents the approximate accumulation interval over which coherent addition remains beneficial before phase misalignment significantly reduces incremental contribution.

From the accumulation behavior, the effective coherence boundary is observed to scale inversely with Doppler frequency:

$$L_c \propto 1/\omega$$

where:

- (L_c) denotes the effective coherence-support width,
- (\omega) represents the Doppler-induced phase rotation rate.

This relationship can also be interpreted through the approximate coherence-collapse condition:

$$\omega L_c \approx \pi$$

which leads to:

$$L_c \approx \pi / \omega$$

The figure demonstrates that:

- low Doppler conditions permit longer coherent accumulation,
- while high Doppler conditions rapidly reduce useful accumulation support.

This inverse relationship is one of the central observations of the study because it connects:

- physical Doppler behavior,
- coherent accumulation limits,
- and adaptive receiver integration control.

The estimated coherence width provides a compact metric that summarizes the useful accumulation region under different channel dynamics.

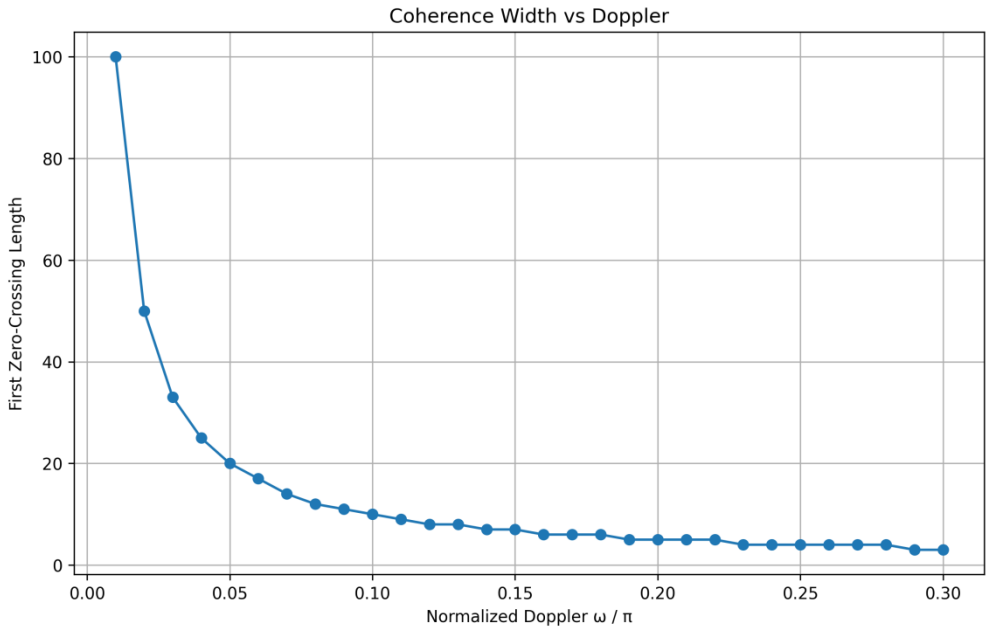
An important implication is that accumulation length should not remain fixed across all channel conditions. Instead, the receiver can potentially adapt its accumulation strategy according to the estimated coherence-support width.

This observation is relevant for:

- synchronization integration control,
- adaptive search-window selection,
- pilot accumulation strategies,
- coherent detector optimization,
- and intelligent PHY-layer adaptation mechanisms.

The coherence-width metric also provides a bridge between:

- theoretical Doppler behavior,
- simulation-based observation,
- and practical receiver adaptation design.



D. Unified Doppler Accumulation Analysis

The previous figures independently examined:

- global correlation accumulation behavior,
- local incremental contribution,
- and coherence-width scaling with Doppler.

This unified figure combines these views into a single representation to visualize the transition from constructive coherent accumulation to destructive accumulation behavior.

The figure simultaneously shows:

- normalized correlation magnitude,
- smoothed incremental gain,
- constructive accumulation region,
- destructive accumulation region,
- and the estimated coherence-support boundary.

The accumulated correlation magnitude is computed as:

```
C(L) = | Σ r[n] s*[n] | ,  n = 0 to L-1
```

while the incremental gain is defined as:

```
g(L) = C_norm(L) - C_norm(L-1)
```

where:

```
C_norm(L) = C(L) / max(C(L))
```

The normalized correlation curve provides the global accumulation trend, while the incremental-gain curve reveals how each additional accumulated sample contributes locally to coherent buildup.

The constructive region corresponds to the interval where:

- incremental gain remains positive,
- indicating that newly accumulated samples continue contributing coherently.

The destructive region begins when:

- incremental gain approaches zero or becomes negative,
- implying that Doppler-induced phase rotation increasingly reduces coherent contribution.

This unified representation provides several important insights:

- Coherent accumulation is not uniformly beneficial across all accumulation lengths.
- The useful accumulation interval shrinks as Doppler increases.
- Local incremental-gain analysis exposes coherence collapse more clearly than correlation magnitude alone.
- The coherence-support boundary emerges naturally from the accumulation dynamics.

The figure also illustrates the close relationship between:

- Doppler-induced phase rotation,
- coherence-support width,
- and adaptive accumulation control.

From a receiver-design perspective, this interpretation suggests that:

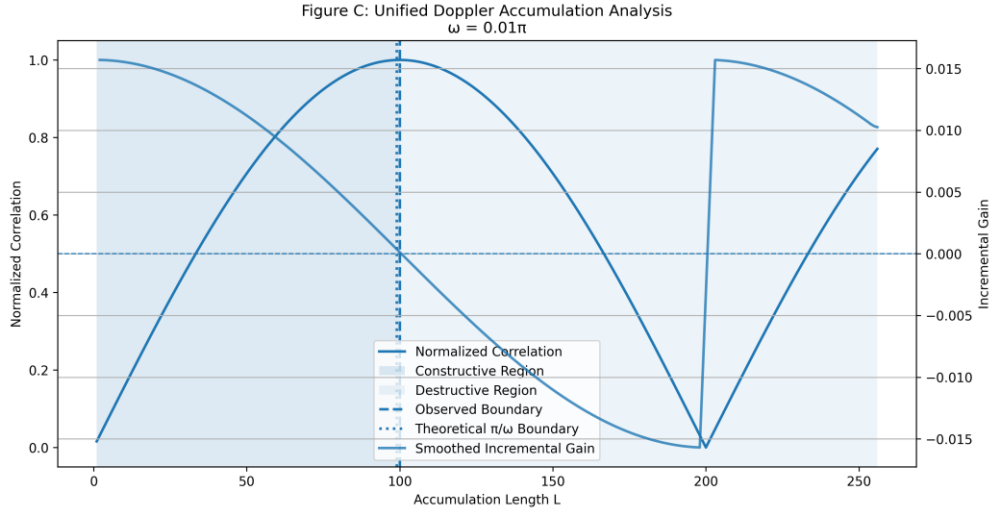
- accumulation length should adapt dynamically to channel time variation,
- unnecessary accumulation beyond the coherence-support region may waste computation and reduce efficiency,
- and local accumulation behavior can provide meaningful adaptation metrics for intelligent PHY-layer control.

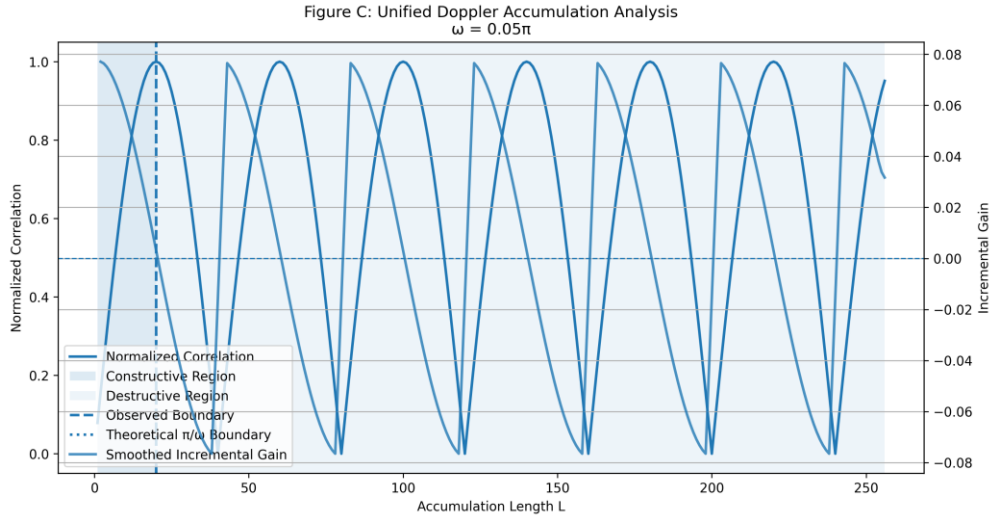
This unified visualization serves as a compact summary of the overall accumulation framework developed in this study and forms the foundation for future extensions involving:

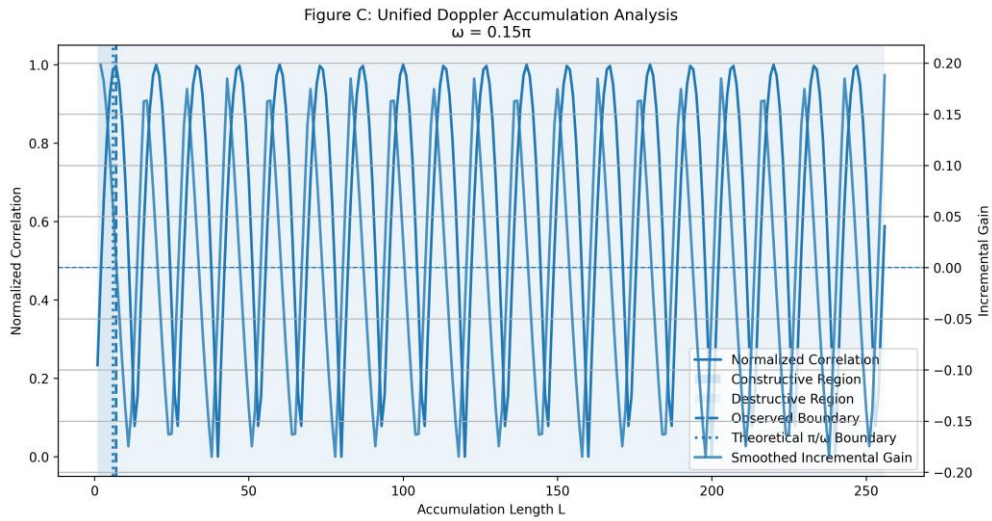
- noisy channels,
- multipath propagation,
- adaptive integration control,

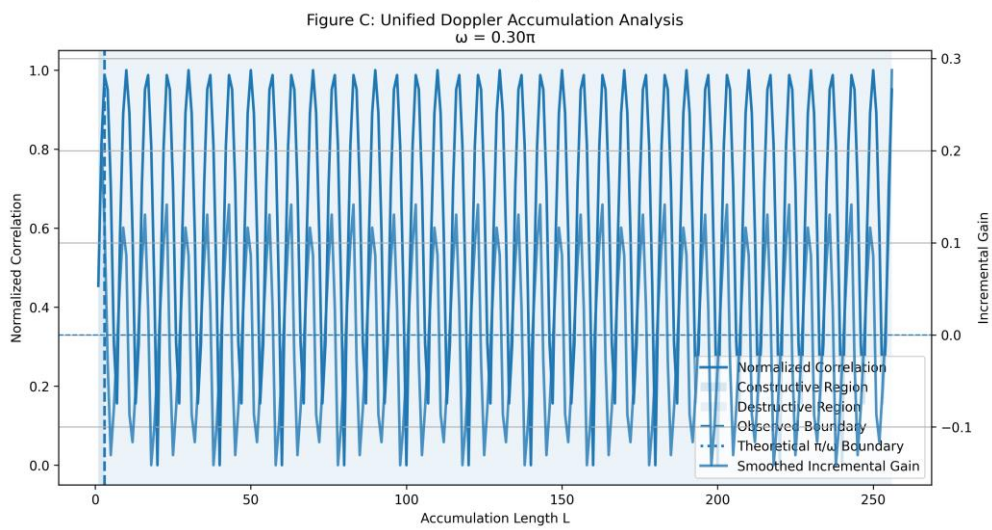
- and AI-assisted coherence estimation.

Normalized correlation









Future Extensions

Figure E — Boundary Estimation Robustness

Planned extension:

- comparison between observed and theoretical coherence boundaries under non-ideal conditions,
- including AWGN, fading, and multipath scenarios.

Potential metric:

```
Boundary Error = |L_observed - L_theory|
```

This analysis will evaluate the robustness of coherence-support estimation under realistic channel impairments.

Figure F — Accumulation Efficiency Analysis

Planned extension:

- analysis of useful versus wasteful accumulation across Doppler conditions.

Potential metrics include:

- useful coherent accumulation fraction,
- constructive/destructive accumulation ratio,
- and adaptive accumulation efficiency versus Doppler.

This extension aims to study Doppler-aware accumulation control and adaptive coherent integration strategies.